

# THE YANG-MILLS EQUATIONS

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## 1 Introduction

Yang-Mills theory is a prime example of a concept that was born out of physical ideas and ambitions, and found its way into the mathematical realm. For physicists, it is the underlying structure of the Standard Model of Particle Physics, one of the greatest achievements in physics so far; for mathematicians, it is a tool, but a powerful one. It was using Yang-Mills theory that Donaldson proved his (Fields Medal winning) results on 4-manifolds, for instance. There is still much that can be done in either direction, and so it seems that this continues to be something worth studying.

We strongly advise the reader that is interested in the interplay between mathematics and physics to look at Yang and Mills' paper [9], as it is only 5 pages long and one can clearly see the expressions for the covariant derivative and the curvature of a connection, without mention to any geometric framework. Indeed, they had no idea of the geometric meaning of their work, nor did they at the time realize how impactful it would be for both physics and mathematics.

To situate the reader, we will first give an introduction to the Maxwell equations, and explain what Yang and Mills were attempting to generalize with their work. Then, we will develop the framework necessary to define the Yang-Mills functional and derive from it the Yang-Mills equation.

This work followed mainly the texts by Atiyah and Bott [1], and Freed and Uhlenbeck [4]. For supplemental details to definitions and proofs not provided in the aforementioned texts and which the author could not easily derive herself, [7] was used.

Throughout the text everything will be taken to be smooth unless stated otherwise.

## 2 Motivation: Maxwell's equations and electromagnetism

Maxwell's equations in their classical form are defined for two vector fields  $E(x, y, z, t)$  and  $B(x, y, z, t)$  on  $\mathbb{R}^3 \times \mathbb{R}$ . For simplicity, we will work with the vacuum equations (where charge and current densities vanish). With the  $\nabla$  operator acting only on the spatial part, i.e.,  $\mathbb{R}^3$ , we have

$$\begin{aligned}\nabla \cdot E &= 0, & \nabla \cdot B &= 0, \\ \nabla \times E &= -\frac{\partial B}{\partial t}, & \nabla \times B &= \mu_0 \epsilon_0 \frac{\partial E}{\partial t}.\end{aligned}$$

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These equations can be derived from a few empirical formulas and some knowledge of vector calculus, but that bears no importance here. Note the similarities on the equations. They are indicative of the more concise form in which the equations can be written, which we shall see ahead.

One important feature of electromagnetism is that we can work with what are called scalar and vector potentials,  $\phi$  and  $A$ , which satisfy

$$\begin{aligned} B &= \nabla \times A, \\ E &= -\nabla\phi - \frac{\partial A}{\partial t}. \end{aligned}$$

Using these potentials, Maxwell's equations written in terms of  $E$  and  $B$  remain unchanged under the transformations

$$\begin{aligned} \phi &\mapsto \phi - \frac{\partial f}{\partial t}, \\ A &\mapsto A + \nabla f, \end{aligned}$$

for some real valued function  $f$ , since  $E$  and  $B$  themselves remain unchanged. This can be checked with a simple computation, remembering that  $\nabla \times \nabla \equiv 0$ . These are called *gauge transformations*, and a choice of  $f$  is referred to as a *choice of gauge*. Specific gauges can be chosen for practical purposes to simplify calculations in different situations. Some famous choices are named, like the Coulomb and Lorenz gauges.

The important thing to keep in mind is that gauge freedom is related to some *arbitrariness* of definition of some quantity. What Yang and Mills were attempting with their paper [9] was to write down field equations which remained invariant under some transformations and conserved the so called isotopic spin, and they relied heavily on an analogy with the electromagnetic case. On the flip side, it can be shown that Maxwell's equations are simply a particular case of the Yang-Mills equations in their current formulation, which we will show in the subsequent sections. To see this, we will need to rewrite the equations in terms of differential forms.

Using an identification of vector fields with forms, we write  $E$  as a 1-form and  $B$  as a 2-form on  $\mathbb{R}^3$ , and so we can rewrite Maxwell's equations with differential forms as<sup>1</sup>

$$\begin{aligned} dB &= 0, & dE - \star \frac{\partial B}{\partial t} &= 0, \\ \star d \star E &= 0, & \star d \star B - \mu_0 \epsilon_0 \frac{\partial E}{\partial t} &= 0. \end{aligned}$$

Let us further rewrite the equations, this time as equations on the Minkowski space<sup>2</sup>  $\mathbb{R}^{3,1}$ . We combine  $E \in \Omega^1(\mathbb{R}^3)$  and  $B \in \Omega^2(\mathbb{R}^3)$  into a 2-form

$$F = B + E \wedge dt \in \Omega^2(\mathbb{R}^{3,1})$$

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<sup>1</sup>The  $\star$  symbol appearing here is the Hodge star operator, which will be properly defined later on. For now it suffices to know that it gives an isomorphism  $\star : \Omega^k(M) \rightarrow \Omega^{n-k}(M)$  for  $M$  an  $n$ -dimensional manifold, and is defined up to a sign on an orthonormal basis  $\{dx^1, \dots, dx^n\}$  by  $\star(dx^1 \wedge \dots \wedge dx^j) = dx^{j+1} \wedge \dots \wedge dx^n$ .

<sup>2</sup>Aside from it being the usual setting for dealing with special relativity, there is nothing special about the Minkowski space for this formulation of Maxwell's equations. We can take spacetime to be any manifold and define an electromagnetic field there as a 2-form  $F$ . However, in order to define separate electric and magnetic fields, we need a splitting  $\mathbb{R} \times S$  for some manifold  $S$  we call space.

known as the electromagnetic field. Thus we have Maxwell's equations in Minkowski space

$$dF = 0, \quad (2.1)$$

$$\star d \star F = 0. \quad (2.2)$$

We would once again like to work with some “vector potential” such that

$$F = dA,$$

however this may only be true locally, and as before  $A$  will not be unique, for a transformation

$$A \mapsto A + df$$

leaves  $F$  unchanged. This is also a gauge transformation. Note that, if we have that  $F$  is exact, then the first equation is a tautology.

*Remark.* While the notion of gauge freedom and arbitrariness might lead the reader to assume that there is no physical meaning in the potentials defined above, that is not necessarily true. If this peaked their interest, we urge the reader to look into the Aharonov-Bohm effect. It is a quantum mechanical effect. We recommend [2, p.130], which gives a brief exposure of the quantum mechanics necessary before explaining the effect.

### 3 Framework: A little bit of gauge theory

Now that we have our motivation and a particular case to look back on and compare, we can go ahead and build the framework. In this section we will give various definitions and remark on several properties of our objects of interest, so that we can properly write down the Yang-Mills equation in the next section. We warn the reader that it may be a bit dry, and ask them to bear with it.

Let  $\pi : P \rightarrow M$  be a principal  $G$ -bundle over an oriented pseudo-Riemannian manifold  $M$  with signature  $(r, s)$ <sup>3</sup>. We call  $G$  the **gauge group**. We denote by  $\tilde{\pi} : \text{Ad}(P) \rightarrow M$  the **associated bundle with fibre  $G$**

$$\text{Ad}(P) := \frac{P \times G}{G},$$

where the action of  $G$  on itself is the adjoint one,  $G \times G \ni (g, h) \mapsto hgh^{-1}$ . Thus we have, for  $p \in \pi^{-1}(x)$ ,

$$\tilde{\pi}^{-1}(x) = \text{Ad}(P)_x = \{[p, g] : g \in G\}.$$

We can identify the sections of  $\text{Ad}(P)$  with the  $G$ -invariant functions  $f : P \rightarrow G$ ,

$$C^\infty(P, G)^G := \left\{ f : P \rightarrow G : f(pg) = g^{-1}f(p)g = \text{Ad}_{g^{-1}}f(p) \right\},$$

such that

- a section  $s \in \Gamma \text{Ad}(P)$  induces a function  $f_s$ , where  $s(\pi(p)) = [p, f_s(p)]$ ,
- a function  $f \in C^\infty(P, G)^G$  induces a section  $s_f(x) = [p, f(p)]$ .

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<sup>3</sup>That is, the metric has  $r$  positive and  $s$  negative signs, so that if  $s = 0$ , the metric is Riemannian.

Note that this does not depend on the choice of  $p \in \pi^{-1}(x)$ , as for any other  $q \in \pi^{-1}(x)$  we have  $q = pg$  for some  $g \in G$ , and so

$$[q, f(q)] = [pg, f(pg)] = [pg, \text{Ad}_{g^{-1}} f(p)] = [p, f(p)]g = [p, f(p)].$$

Furthermore, the sections of  $\text{Ad}(P)$  form a group under pointwise multiplication,

$$(s \cdot s')(x) = s(x)s'(x) = [p, f(p)f'(p)].$$

The  $G$ -invariant functions can further be identified with the **G-bundle automorphisms of  $P$** , that is, fibre preserving diffeomorphisms  $u : P \rightarrow P$  that are  $G$ -equivariant,

$$u(pg) = u(p)g.$$

We denote this group by  $\text{Aut}(P)$ . This identification can be written as follows:

- A function  $f \in C^\infty(P, G)^G$  induces an automorphism  $\tilde{f} : P \rightarrow P$ ,  $\tilde{f}(p) = pf(p)$ , which is clearly fibre preserving, and equivariant,

$$\tilde{f}(pg) = pgf(pg) = pf(p)g = \tilde{f}(p)g.$$

- Conversely, there is a unique map  $f : P \rightarrow G$  such that  $u(p) = pf(p)$  for a given  $u \in \text{Aut}(P)$ . The equivariance of  $u$  gives

$$u(pg) = p(gf(pg)) = p(f(p)g) = u(p)g,$$

and so  $gf(pg) = f(p)g$ , which establishes  $f \in C^\infty(P, G)^G$ .

Thus we have the group isomorphisms

$$\Gamma \text{Ad}(P) \simeq C^\infty(P, G)^G \simeq \text{Aut}(P),$$

and we will call an element of  $\mathcal{G} := \text{Aut}(P)$  a **gauge transformation**.<sup>4</sup> It can be shown that  $\mathcal{G}$  is an infinite dimensional Lie group, but we will not attempt to prove this here. We will, however, work with its Lie algebra.

Let  $\mathfrak{g}$  be the Lie algebra of  $G$ , and let  $G$  act on  $\mathfrak{g}$  via the adjoint action

$$g \mapsto d_e(\text{Ad}_g) : \mathfrak{g} \rightarrow \mathfrak{g}.$$

Then  $\text{ad}(P) \rightarrow M$  is an **associated vector bundle with standard fibre  $\mathfrak{g}$** ,

$$\text{ad}(P) := \frac{P \times \mathfrak{g}}{G}.$$

Just as with  $\text{Ad}(P)$ , we can identify the sections of  $\text{ad}(P)$  with  $G$ -invariant functions  $P \rightarrow \mathfrak{g}$ ,

$$C^\infty(P, \mathfrak{g})^G := \left\{ f : P \rightarrow \mathfrak{g} : f(pg) = g^{-1}f(p) \right\},$$

such that any section can be written  $s_f(x) = [p, f(p)]$ , for any  $p \in \pi^{-1}(x)$ .

The space of sections  $\Gamma \text{ad}(P)$  has a natural Lie algebra structure induced by the bracket  $\{-, -\}$  on  $\mathfrak{g}$ ,

$$\{s, s'\}(x) = \{[p, f(p)], [p, f'(p)]\} = [p, \{f(p), f'(p)\}].$$

We affirm then that this is the Lie algebra of  $\mathcal{G}$ .

Now we would like to introduce a metric (i.e., an inner product) on  $\Omega^k(M, \text{ad}(P))$ . It arises naturally given a (pseudo) Riemannian metric on  $M$  and a metric on  $\mathfrak{g}$ . First, we need a result that we will not prove, see [6, Prop. 4.24] for a proof.

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<sup>4</sup>A word of caution: some call  $\mathcal{G}$  the gauge group instead of  $G$ .

**Lemma.** *Let  $G$  be a compact Lie group. Then the Lie algebra  $\mathfrak{g}$  admits an inner product which is invariant under the adjoint action of  $G$  on  $\mathfrak{g}$ .*

In view of this result, from now on  $G$  will be a compact Lie group. Then, letting  $\langle -, - \rangle_{\mathfrak{g}}$  be such an inner product on  $\mathfrak{g}$ , we induce a fibre metric  $h$  on  $\text{ad}(P)$ ,

$$h(s, s')(x) = h_x(s(x), s'(x)) = h_x([p, f_s(p)], [p, f_{s'}(p)]) := \langle f_s(p), f_{s'}(p) \rangle_{\mathfrak{g}}.$$

Making use of the Riemannian structure on  $M$ , we can define the Hodge star operator

$$\star : \Omega^k(M) \rightarrow \Omega^{n-k}(M)$$

as the only map that satisfies

$$\alpha \wedge \star \beta = \langle \alpha, \beta \rangle_M \text{vol},$$

where  $\langle -, - \rangle_M$  is the inner product on  $M$ , given by the metric  $g$ , and  $\text{vol}$  is the volume form on  $M$ . Then, noting that  $\alpha \wedge \star \beta$  is a top-form, we can define a metric on  $\Omega^k(M)$ :

$$\langle \alpha, \beta \rangle_{L^2} := \int_M \langle \alpha, \beta \rangle_M \text{vol} = \int_M \alpha \wedge \star \beta.$$

Similarly, we can extend the **Hodge star** to  $\Omega^k(M, \text{ad}(P))$  and use the fibre metric  $h$  to induce a metric on  $\Omega^k(M, \text{ad}(P))$ . Given a point  $x \in M$ , choose a local frame  $s$  of  $\text{ad}(P)$  on a neighbourhood  $U \subseteq M$  of  $x$ . Then for  $\alpha \in \Omega^k(M, \text{ad}(P))$  and  $\beta \in \Omega^l(M, \text{ad}(P))$ , we can write  $\alpha = \alpha^i \otimes s_i$  and  $\beta = \beta^j \otimes s_j$ , where  $\alpha^i, \beta^j \in \Omega^{\bullet}(M)$ . Then  $h$  induces a pairing

$$\dot{\wedge} : \Omega^k(M, \text{ad}(P)) \times \Omega^l(M, \text{ad}(P)) \rightarrow \Omega^{k+l}(M),$$

$$(\alpha \dot{\wedge} \beta)_x := (\alpha^i \wedge \beta^j)_x h_x(s_i(x), s_j(x)),$$

and finally extending the Hodge star

$$\star \alpha := (\star \alpha^i) \otimes s_i,$$

we have that  $\alpha \dot{\wedge} \star \beta$  is an  $n$ -form on  $M$ , for any  $\alpha, \beta \in \Omega^k(M, \text{ad}(P))$ , and so we can finally define

$$\langle \alpha, \beta \rangle_{L^2} := \int_M \alpha \dot{\wedge} \star \beta,$$

the **metric on  $\Omega^k(M, \text{ad}(P))$**  we were seeking.

Just as we can use the Hodge star to define the **dual**  $d^* : \Omega^k(M) \rightarrow \Omega^{k-1}(M)$  by

$$\langle d^* \alpha, \beta \rangle_{L^2} := \langle \alpha, d\beta \rangle_{L^2},$$

we can define the dual  $d_{\nabla}^* : \Omega^k(M, \text{ad}(P)) \rightarrow \Omega^k(M, \text{ad}(P))$  with

$$\langle \alpha, d_{\nabla}^* \beta \rangle_{L^2} := \langle d_{\nabla} \alpha, \beta \rangle_{L^2},$$

for a connection  $\nabla$  on  $\text{ad}(P)$ . Note that for a  $k$ -form  $\alpha$  we have

$$d_{\nabla}^* \alpha = (-1)^{n(k-1)+s+1} \star d \star \alpha,$$

where  $n$  is the dimension of  $M$  and  $s$  comes from its signature as a pseudo-Riemannian manifold, and is  $s = 0$  if  $M$  is Riemannian.

Next, we need to talk a little about connections. A **connection** on  $P$  can be defined in many forms; the definition we will use here is that of a 1-form  $\omega \in \Omega^1(P, \mathfrak{g})$  such that it satisfies

$$(i) \quad \omega_{pg}(vg) = g^{-1}\omega_p(v)g = \text{Ad}_{g^{-1}}\omega, \quad \forall v \in T_pP, \quad g \in G,$$

$$(ii) \quad \omega_p(p\xi) = \xi, \quad \forall \xi \in \mathfrak{g},$$

that is, it is  $G$ -invariant and takes fixed values on vertical tangent vectors. Here, we use the convention

$$p\xi := \left. \frac{d}{dt} \right|_{t=0} p \cdot \exp(t\xi) \in T^V P$$

for the infinitesimal action of  $\mathfrak{g}$  on  $P$ . Thus, at times  $\xi \in \mathfrak{g}$  will also denote the field generated by  $\xi$  on  $P$ . We shall denote the set of smooth connections on  $P$  by  $\mathcal{A}$ .

If we choose a fixed connection  $\tilde{A} \in \mathcal{A}$ , then we can write any connection as the sum of  $\tilde{A}$  and a basic<sup>5</sup>  $\mathfrak{g}$ -valued 1-form, i.e.,

$$\mathcal{A} = \tilde{A} + \Omega_{bas}^1(P, \mathfrak{g})$$

is an affine space modelled after the vector space  $\Omega_{bas}^1(P, \mathfrak{g})$ .

Then, since we have an isomorphism

$$\Omega_{bas}^k(P, \mathfrak{g}) \simeq \Omega^k(M, \text{ad}(P)),$$

for a given fixed connection  $\tilde{A}$ , we can see the curvature  $F(\omega)$  of a connection on  $P$ , which is given by the structure equation

$$F(\omega) = d\omega + \frac{1}{2}[\omega, \omega],$$

as a 2-form  $F(\omega) \in \Omega^2(M, \text{ad}(P))$ .

## 4 The Yang-Mills Equation

Finally, we have all the necessary structure to write down the **Yang-Mills functional**, using the inner product on  $\Omega^k(M, \text{ad}(P))$  we have defined. For a connection  $A$  on  $P$ , we have

$$YM(A) = \langle F(A), F(A) \rangle_{L^2} = \int_M F(A) \wedge \star F(A).$$

The Yang-Mills equations will simply be conditions that a connection on  $P$  must satisfy in order to represent a stationary point of this functional. So, we wish to see what happens to  $YM(A)$  when we vary  $A$ . Since  $\mathcal{A}$  is an affine space, it suffices to look at

$$\delta_B YM(A) = \left. \frac{d}{dt} \right|_{t=0} YM(A + tB) = 0,$$

for any  $B \in \Omega^1(M, \text{ad}(P))$ . Using the structure equation, we have

$$F(A + tB) = F(A) + td_AB + \frac{t^2}{2}[B, B],$$

and thus

$$\begin{aligned} \delta_B YM(A) &= \left. \frac{d}{dt} \right|_{t=0} \langle F(A) + td_AB + \frac{t^2}{2}[B, B], F(A) + td_AB + \frac{t^2}{2}[B, B] \rangle \\ &= 2\langle F(A), d_AB \rangle \\ &= 2\langle d_A^* F(A), B \rangle = 0. \end{aligned}$$

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<sup>5</sup>We define a differential form to be **basic** if it is  $G$ -invariant and kills vertical tangent vectors.

Since we want this to be true for arbitrary  $B$ , we must have

$$d_A^* F(A) = \star d_A \star F(A) = 0,$$

for a stationary connection for the Yang-Mills functional. *This* is the **Yang-Mills equation**. A connection which satisfies this equation is called a **Yang-Mills connection**.

Going back now and comparing this to Maxwell's equations as we wrote them, one can see that this corresponds to Equation 2.2. Equation 2.1 is also valid here, and is simply the **Bianchi identity**

$$d_A F(A) = 0.$$

Electromagnetism is simply a Yang-Mills theory with gauge group  $U(1)$ .

The situation we are most interested in is when  $M$  is a 4-dimensional manifold, as it allows for the notion of **self-dual** and **anti-self-dual** connections, that is, connections whose curvature satisfies, respectively,

$$F(A) = \pm \star F(A).$$

Note that as a consequence of the Bianchi equation, self-dual or anti-self dual connections automatically satisfy the Yang-Mills equation. A solution of this form is called an **instanton**, and the space

$$\mathcal{M} = \{A \in \mathcal{A} : F(A) = -\star F(A)\} / \mathcal{G},$$

is called the **moduli space of instantons**.

## 5 Applications and interesting results

As was mentioned in the introduction, Yang-Mills theory has proven to be a very useful tool in mathematics. Perhaps most famous is Donaldson's result on the topology of 4-manifolds, for which he made use of the topology of  $\mathcal{M}$  on a principal  $SU(2)$  bundle.

**Definition.** The intersection form  $\omega$  on a compact, simply connected, smooth 4-manifold  $M$  is defined using the de Rham cohomology. Let  $\alpha, \beta \in H_{dR}^2(M)$  be represented by 2-forms  $\alpha, \beta \in \Omega^2(M)$ , then

$$\omega(\alpha, \beta) = \int_M \alpha \wedge \beta.$$

**Theorem** (Donaldson, 1983). *Let  $M$  be a compact, simply connected, smooth 4-manifold, and let  $\omega$  be the intersection form on  $M$ . Then, if  $\omega$  is positive-definite, it is diagonalisable over the integers to the identity matrix.*

This result, along with work by Freedman on the classification of topological 4-manifolds, implies the following.

**Corollary.** *There exist non-smoothable topological 4-manifolds.*

There is an even more striking result which follows from Donaldson's theorem. A **fake**  $\mathbb{R}^4$  is a 4-manifold which is homeomorphic but not diffeomorphic to the usual  $\mathbb{R}^4$ .

**Theorem** (Taubes, 1987). *There is an uncountable collection of pairwise non-diffeomorphic fake  $\mathbb{R}^4$ s.*

A detailed proof of Donaldson's theorem can be found in book [4], which was written mainly for this purpose. The book [3] also provides a proof to this theorem, but more generally it deals with applications of Yang-Mills theory to 4-manifold topology. An interesting result in the context of integrability is the Koszul-Malgrange theorem [3, Thm. 2.1.53], which can be seen as a special case of the Newlander-Nirenberg theorem for integrable almost complex structures.

Moreover, one of the famous Millennium Prize Problems concerns Yang-Mills theory, in the context of Quantum Field Theory. We will not venture to explain the problem, since it would lead to a discussion of quantisation which is not within the scope of this work. In the text [5], there is a concise explanation of the physics and mathematics surrounding Yang-Mills, the possible mathematical implications, and the statement of the problem itself, which goes as follows.

**Open Problem** (Yang-Mills Existence and Mass Gap). *Prove that for any compact simple gauge group  $G$ , a non-trivial quantum Yang-Mills theory exists on  $\mathbb{R}^4$  and has a mass gap  $\Delta > 0$ .*

Finally, the Yang-Mills equations are partial differential equations, and there has been extensive work done on the analytic side of Yang-Mills theory. The work done here represents the beginning of a master's project, which has as its main objective an understanding of the proof of the Uhlenbeck compactness theorem [8], an analytic result which was used by Donaldson in the proof of his theorem. It can be roughly stated as follows<sup>6</sup>.

**Theorem** (Uhlenbeck, 1982). *Every sequence of Yang-Mills connections with uniformly bounded curvature is gauge equivalent to a sequence which has a  $C^\infty$ -convergent subsequence.*

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<sup>6</sup>The theorem stated is known as *strong* Uhlenbeck compactness. There is also *weak* Uhlenbeck compactness, which deals with sequences of connections in general, and yields weak  $W^{1,p}$  convergence.

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